1.1.3 - Spring Mass Damper System

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March 25, 2026

Question

A spring-mass-damper system is excited by a force $F_0 \sin \omega t$. The amplitude at resonance is measured to be 1 cm. At half the resonance frequency, the amplitude is 0.5 cm. The damping ratio of the system is:

a) 0.1026

b) 0.3242

c) 0.7211

d) 0.1936

Theoretical Solution

From Newton's Second Law:

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin(\omega t)$$

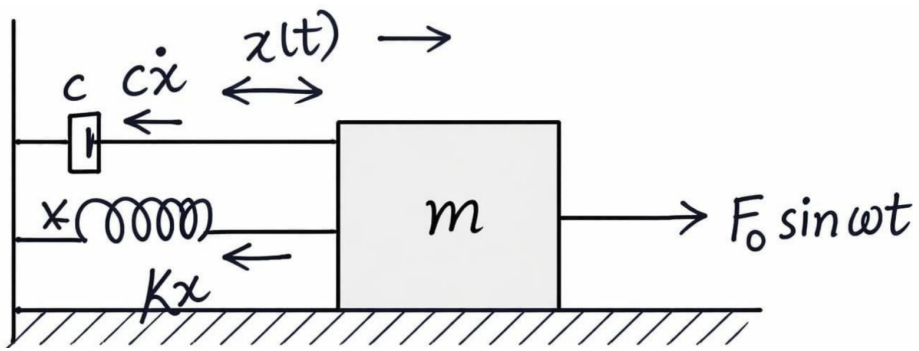

Figure: Spring Mass Damper System

System Response

For a second-order differential equation, two functions $x_1(t)$ and $x_2(t)$ that are linearly independent statisfy the DE.

$$x(t) = x_h(t) + x_p(t)$$

$x_h(t) \rightarrow$ transient response(complementary solution)

$x_p(t) \rightarrow$ steady-state response(particular solution)

Assumed Solution

Consider the homogeneous equation:

$$m\ddot{x} + c\dot{x} + kx = 0 \tag{1}$$

Assume an exponential solution:

$$x(t) = e^{st} \tag{2}$$

Then:

$$\dot{x} = se^{st}, \quad \ddot{x} = s^2 e^{st} \tag{3}$$

Characteristic Equation

Substitute into the differential equation:

$$ms^2e^{st} + cse^{st} + ke^{st} = 0 \tag{4}$$

Divide by e^{st}:

$$ms^2 + cs + k = 0 \tag{5}$$

Roots of Characteristic Equation

$$s = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m} \tag{6}$$

Define:

$$\omega_n = \sqrt{\frac{k}{m}}, \quad \zeta = \frac{c}{2\sqrt{km}} \tag{7}$$

General Solution

$$x_h(t) = C_1 x_1(t) + C_2 x_2(t) \tag{8}$$

$$x_h(t) = C_1 e^{(-\zeta\omega_n + i\omega_d)t} + C_2 e^{(-\zeta\omega_n - i\omega_d)t} \tag{9}$$

Convert to Real Form

Factor:

$$x_h(t) = e^{-\zeta\omega_n t}\left(C_1 e^{i\omega_d t} + C_2 e^{-i\omega_d t}\right) \tag{10}$$

Using Euler's formula:

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{11}$$

$$x_h(t) = e^{-\zeta\omega_n t}\left(A\cos\omega_d t + B\sin\omega_d t\right) \tag{12}$$

Physical Interpretation

$e^{-\zeta\omega_n t} \rightarrow$ exponential decay

$\cos, \sin \rightarrow$ oscillatory motion

$\omega_d < \omega_n \rightarrow$ reduced frequency due to damping

Physical Meaning of Transient

Caused by initial conditions

Energy dissipates due to damping

$$x(t) \sim e^{-\zeta \omega_n t}$$

Eventually vanishes

Laplace Transform

$$ms^2X(s) + csX(s) + kX(s) = \frac{F_0\omega}{s^2+\omega^2}$$

$$X(s) = \frac{F_0\omega}{(s^2+\omega^2)(ms^2+cs+k)}$$

Using Partial Fractions Expansion:

$$\frac{F_0\omega}{(s^2+\omega^2)(ms^2+cs+k)} = \frac{As+B}{s^2+\omega^2} + \frac{Cs+D}{ms^2+cs+k}$$

Inverse Laplace Transform

Applying $\mathcal{L}^{-1}$ to the steady-state term $\dfrac{A}{s - i\omega}$:

$$\mathcal{L}^{-1}\left\{\frac{A}{s - i\omega}\right\} = A\, e^{i\omega t}$$

Substituting $A = \dfrac{F_0}{(k - m\omega^2) + ic\omega}$:

$$x_{ss}(t) = \frac{F_0}{(k - m\omega^2) + ic\omega} \cdot e^{i\omega t}$$

Writing the denominator in polar form $re^{i\phi}$ where:

$$r = \sqrt{(k - m\omega^2)^2 + (c\omega)^2}, \quad \phi = \arctan\left(\frac{c\omega}{k - m\omega^2}\right)$$

Final Steady-State Response

Therefore:

$$x_{ss}(t) = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}} \cdot e^{i(\omega t - \phi)}$$

Taking the **real part** for a cosine excitation:

$$\boxed{x_{ss}(t) = X\cos(\omega t - \phi)}$$

where:

$$X = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

$$\phi = \arctan\left(\frac{c\omega}{k - m\omega^2}\right)$$

Complete Solution

$$x(t) = e^{-\zeta\omega_n t}(A\cos\omega_d t + B\sin\omega_d t) + X\sin(\omega t - \phi)$$

As $t \to \infty$:

$$e^{-\zeta\omega_n t} \to 0$$

$$x(t) \to X\sin(\omega t - \phi)$$

Non-Dimensional Form

Define:

$$\omega_n = \sqrt{\frac{k}{m}}, \quad r = \frac{\omega}{\omega_n}, \quad \zeta = \frac{c}{2\sqrt{km}} \tag{13}$$

$$X = \frac{F_0/k}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} \tag{14}$$

Apply Conditions

At resonance $r \approx 1$, As frequency of oscillation would be equal to natural frequency,

$$X_{\text{res}} = \frac{F_0/k}{2\zeta} = 1$$

$$X_{0.5} = \frac{F_0/k}{\sqrt{0.75^2 + \zeta^2}} = 0.5$$

Solve

$$\frac{2\zeta}{\sqrt{0.75^2 + \zeta^2}} = 0.5$$

$$\zeta = 0.1936$$

Final Answer

$$\boxed{\zeta = 0.1936}$$

Correct option: **(d)**

Python Plot: Displacement vs Time

```
import numpy as np
import matplotlib.pyplot as plt
# System parameters
zeta = 0.1936
omega_n = 1.0
omega = omega_n
# Derived quantities
omega_d = omega_n * np.sqrt(1 - zeta**2)
# Constants
A = 1.0
B = 0.0
# Steady-state amplitude
X = 1 / (2 * zeta)
```

Python Plot: Displacement vs Time

```
# Phase at resonance
phi = np.pi / 2
# Time range (increased)
t = np.linspace(0, 150, 4000)
# Transient response
x_h = np.exp(-zeta * omega_n * t) * \
        (A*np.cos(omega_d*t) + B*np.sin(omega_d*t))

# Steady-state response
x_p = X * np.sin(omega * t - phi)
# Total response
x = x_h + x_p
# Plot
plt.plot(t, x, label="Total")
plt.plot(t, x_h, '--', label="Transient")
plt.plot(t, x_p, ':', label="Steady-State")
```

Python Plot: Displacement vs Time

```
plt.xlabel("Time")
plt.ylabel("Displacement")
plt.title("Transient to Steady−State Response")
plt.legend()
plt.grid()
plt.show()
```

Plot

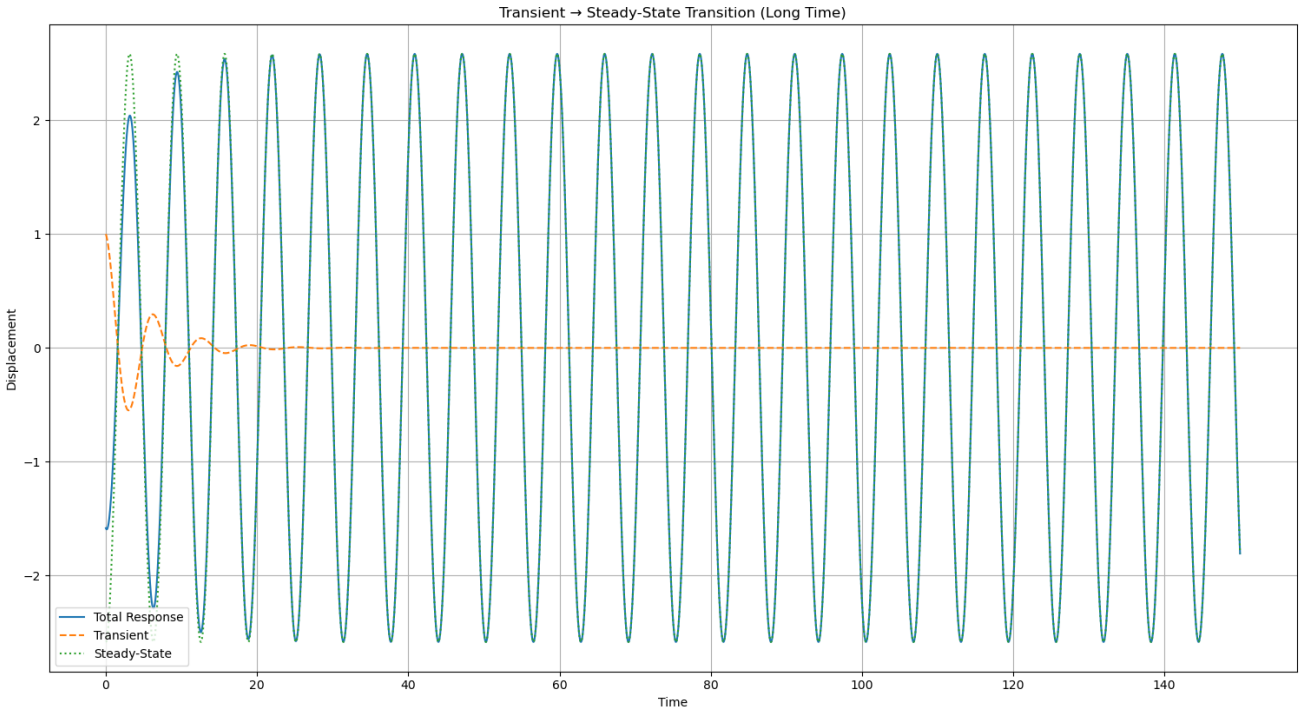

Figure: Plot of Displacement versus Time